

Post-processing in LSSS: Korona modules

Cleaning and automatic categorisation of multi-frequency acoustic data, with examples from krill and capelin surveys

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What we will cover today

OUTLINE

01

The Korona module

What Korona can do: cleaning data and categorising backscatter automatically

02

Acoustic categorisation

Echograms, frequency response, why several frequencies separate animal groups, and recent research

03

Case studies from MFRI

Krill around Iceland in spring; krill and capelin in East Greenland in autumn

04

Hands-on with LSSS libraries

Running Korona, adding categories and checking the output on your own data

Course material: [.../u5/bergmal/LSSS_namskeid/MFRI_Advance_acoustic_workshop_2026](https://github.com/u5/bergmal/LSSS_namskeid/MFRI_Advance_acoustic_workshop_2026)

What the Korona module can do

01 · KORONA MODULE

Korona is the batch-processing program that comes with LSSS. It runs a chain of modules on the raw multi-frequency data, so that cleaning and categorisation are **reproducible** from survey to survey.

Clean the data

- Remove air bubbles near the surface
- Remove spikes and impulse noise
- Remove background noise at depth

Categorise backscatter

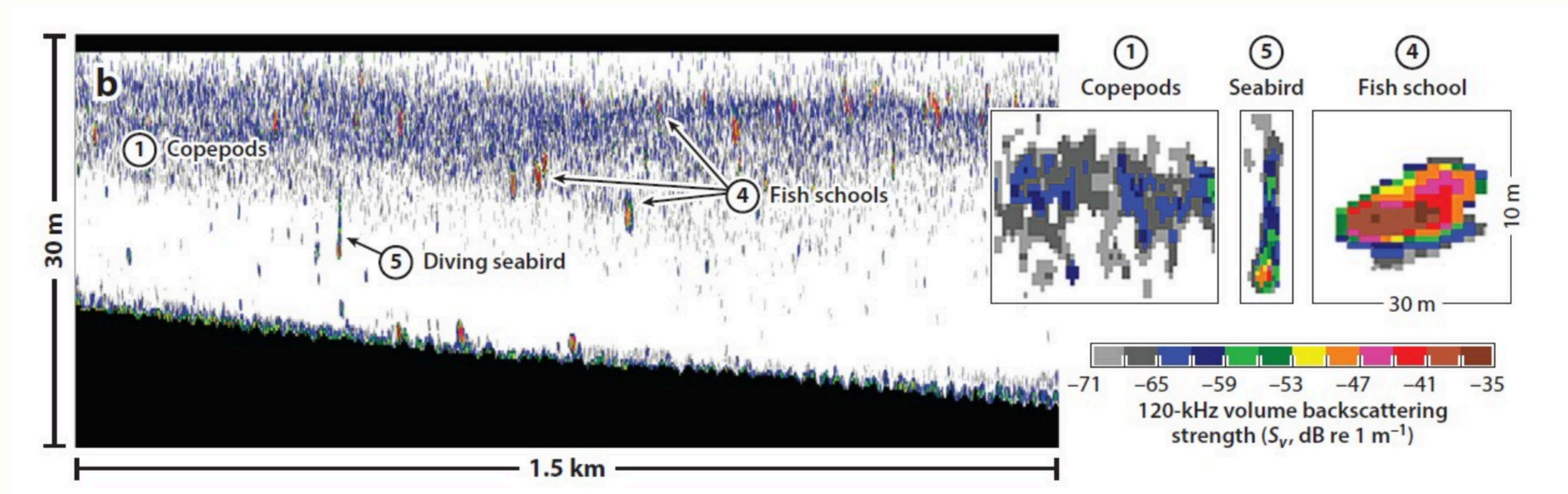
- Into species or acoustic groups
- Detect and classify schools
- Separate plankton groups

Feed the survey

- NASC per category, ready for scrutiny in LSSS
- Same settings applied every year

Reading an echogram

02 · ACOUSTIC CATEGORISATION



Copepods: weak, diffuse scattering layers

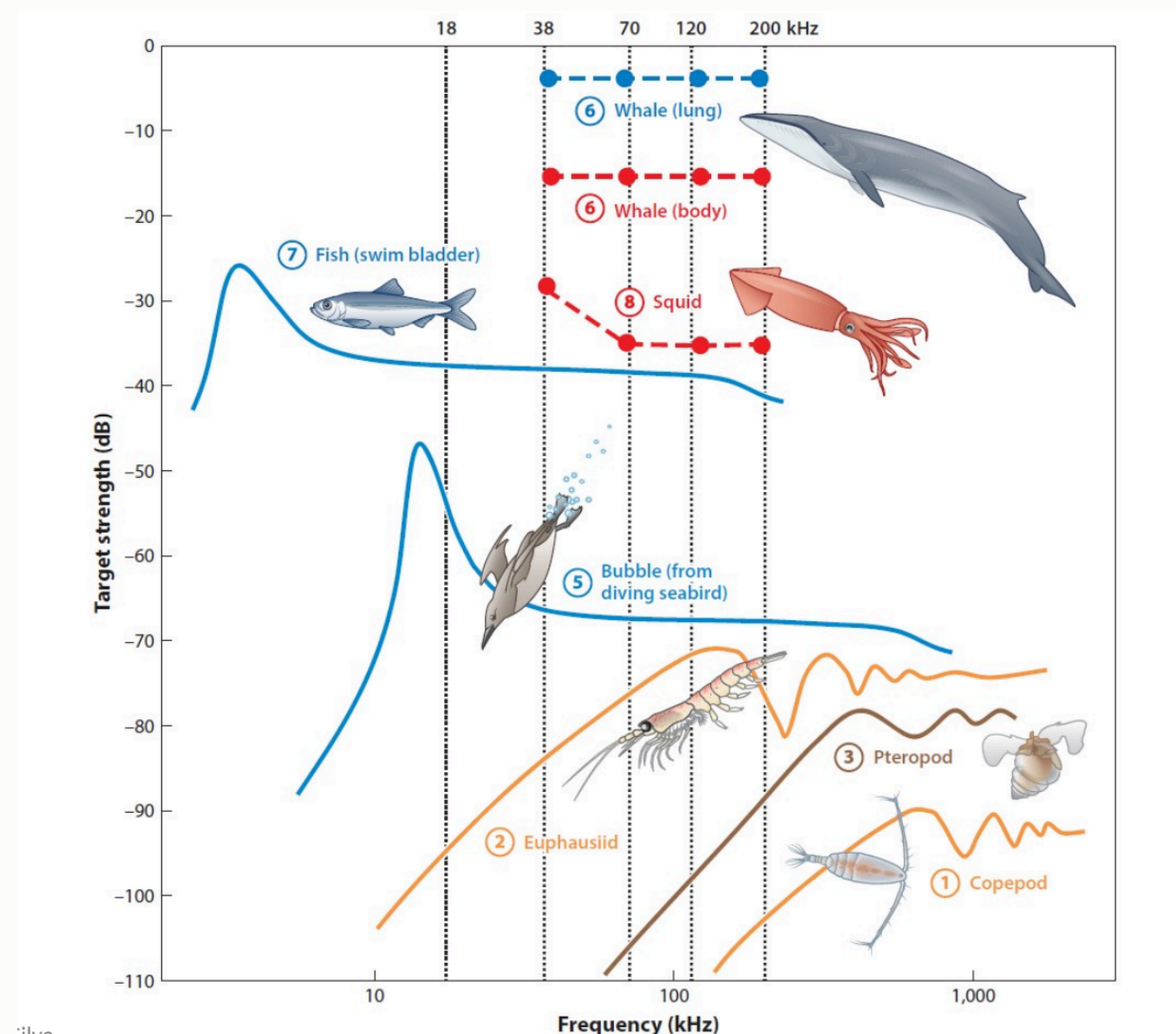
Diving seabird: vertical trace of the bird and its air bubbles

Fish schools: dense, discrete patches with strong S_v

Figure: Benoit-Bird & Lawson (2016)

Frequency response of different animals

02 · ACOUSTIC CATEGORISATION



Target strength depends on frequency and on the animal's **size, shape and material**.

- **Fish with a swimbladder:** the gas gives a strong, fairly flat response across the echosounder frequencies.
- **Fluid-like zooplankton** (krill, copepods): weak at low frequencies, rising steeply as frequency increases.

Figure: Benoit-Bird & Lawson (2016)

Krill echo more strongly at 120 kHz than at 38 kHz. Fish with a swimbladder do not.

KEY PRINCIPLE

This difference in backscatter between frequencies is the signal Korona uses to separate krill from fish.

What is new: 2025–2026

02 · ACOUSTIC CATEGORISATION · RECENT RESEARCH

Korona in the Central Arctic Ocean

Korona, with machine-learning elements, removed noise and assigned backscatter to taxonomic groups of fish and plankton on surveys in 2022 and 2023 with R/V *Kronprins Haakon*, reaching the North Pole in 2022.

Misund et al. (2025), *Sensors*

Deep learning for krill

A U-Net network extracted Antarctic krill from EK60 data. Three frequencies (38, 70 and 120 kHz) worked best, with an F1 score of about 98 %, and 120 kHz contributed most.

Li et al. (2025), *Mar. Ecol. Prog. Ser.*

What still holds AI back

Gains from deep learning on echosounder data have been modest so far. The main barriers are the lack of standard data formats and of high-quality labelled data.

Malde (2026), arXiv preprint

Take-home: the multi-frequency contrast is still the core signal, and carefully scrutinised Korona libraries are exactly the labelled data that new methods need.

Krill around Iceland on the spring survey

03 · CASE STUDY 1

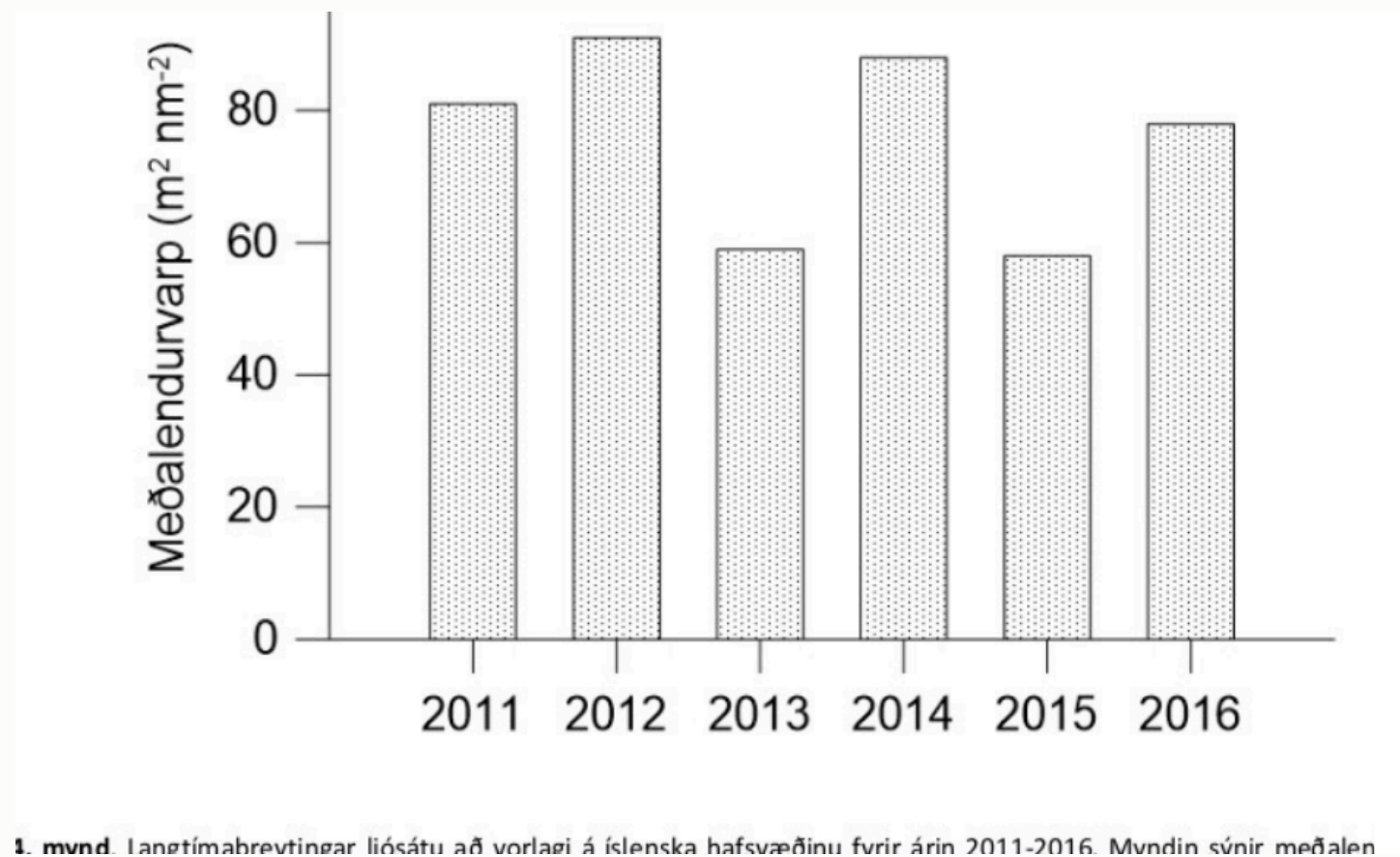
From manual scrutiny to automatic categorisation in Korona, 2011–2023



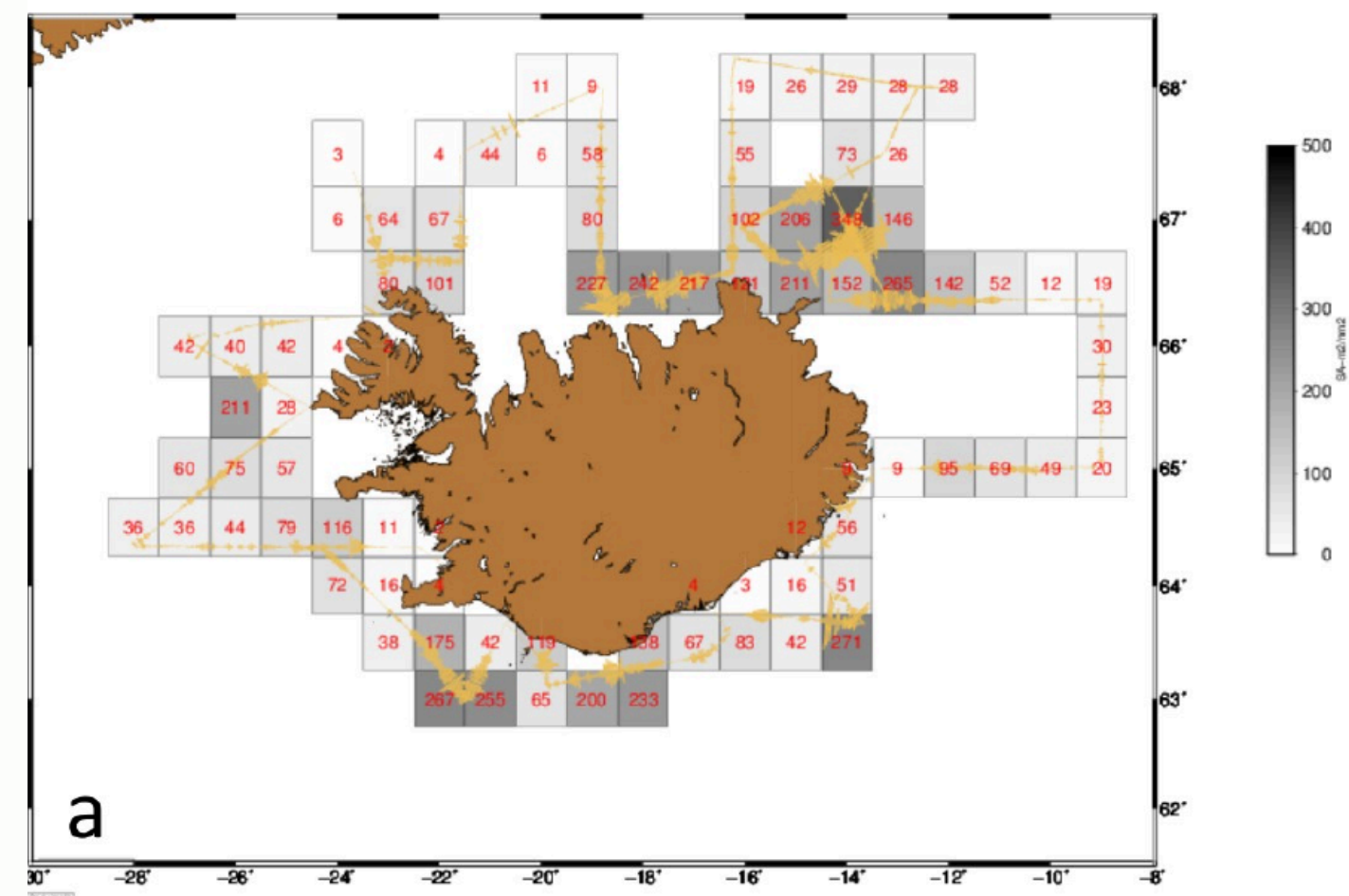
Krill assessment on the spring survey

03 · CASE STUDY 1 · KRILL AROUND ICELAND

Krill backscatter has been measured on the Icelandic spring survey since 2011. In the last published MFRI summary (2011–2016), annual mean NASC varied between about 60 and 90 $\text{m}^2 \text{nmi}^{-2}$, with no clear trend.



Annual mean NASC ($\text{m}^2 \text{nmi}^{-2}$), 2011–2016



Mean NASC per statistical square

Figures: Ólafsdóttir et al. (2018), Ástand sjávar, MFRI

Methodology

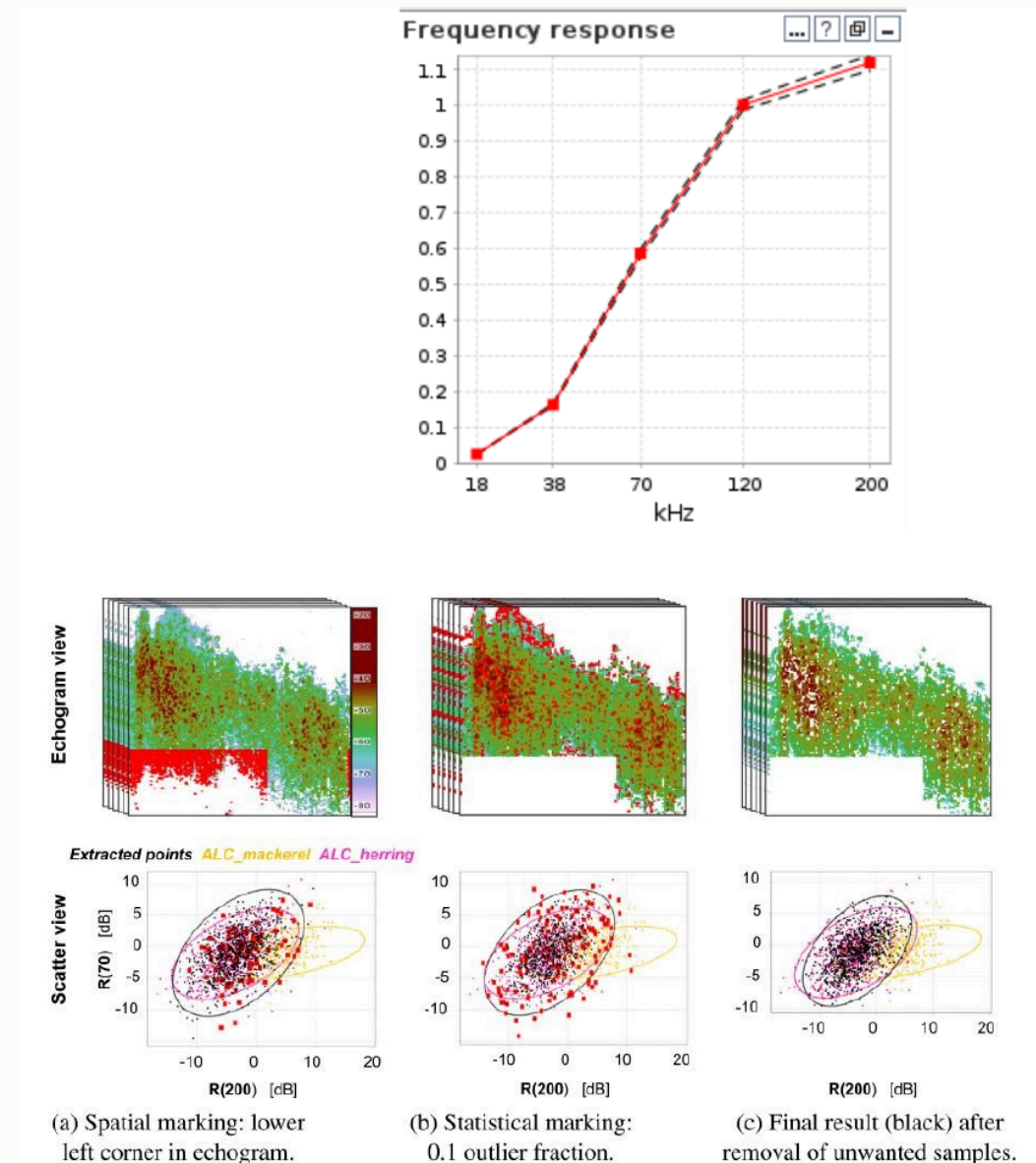
03 · CASE STUDY 1 · KRILL AROUND ICELAND

Acoustic measurements of krill

Spring survey around Iceland, 0–300 m depth. The backscatter difference between 38 and 120 kHz proved effective for distinguishing krill from fish.

Procedure in LSSS

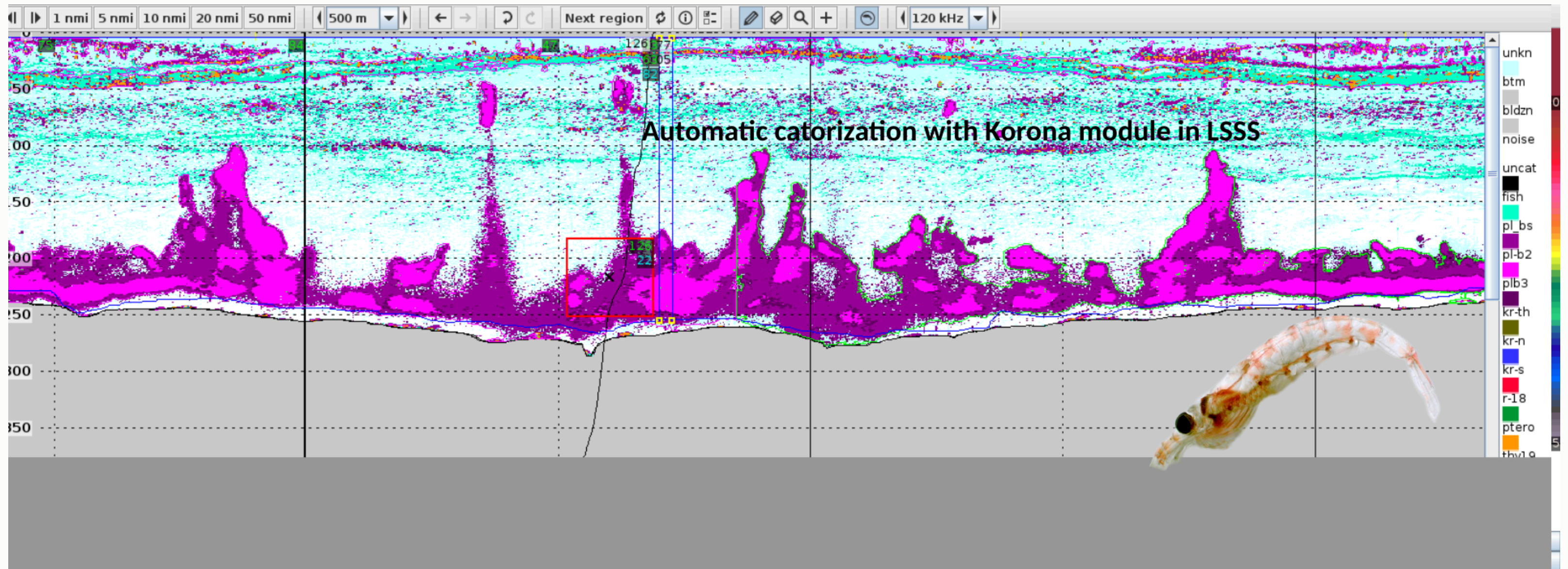
1. Scrutinise echograms: remove noise, bad data and bottom
2. Identify krill and other categories on echograms
3. Biological sampling, 0–300 m
4. Build an acoustic feature library
5. Categorise krill automatically in Korona



Reynisson & Gíslason (2015) · Figure: Korneliussen et al. (2016)

Automatic selection of krill backscatter

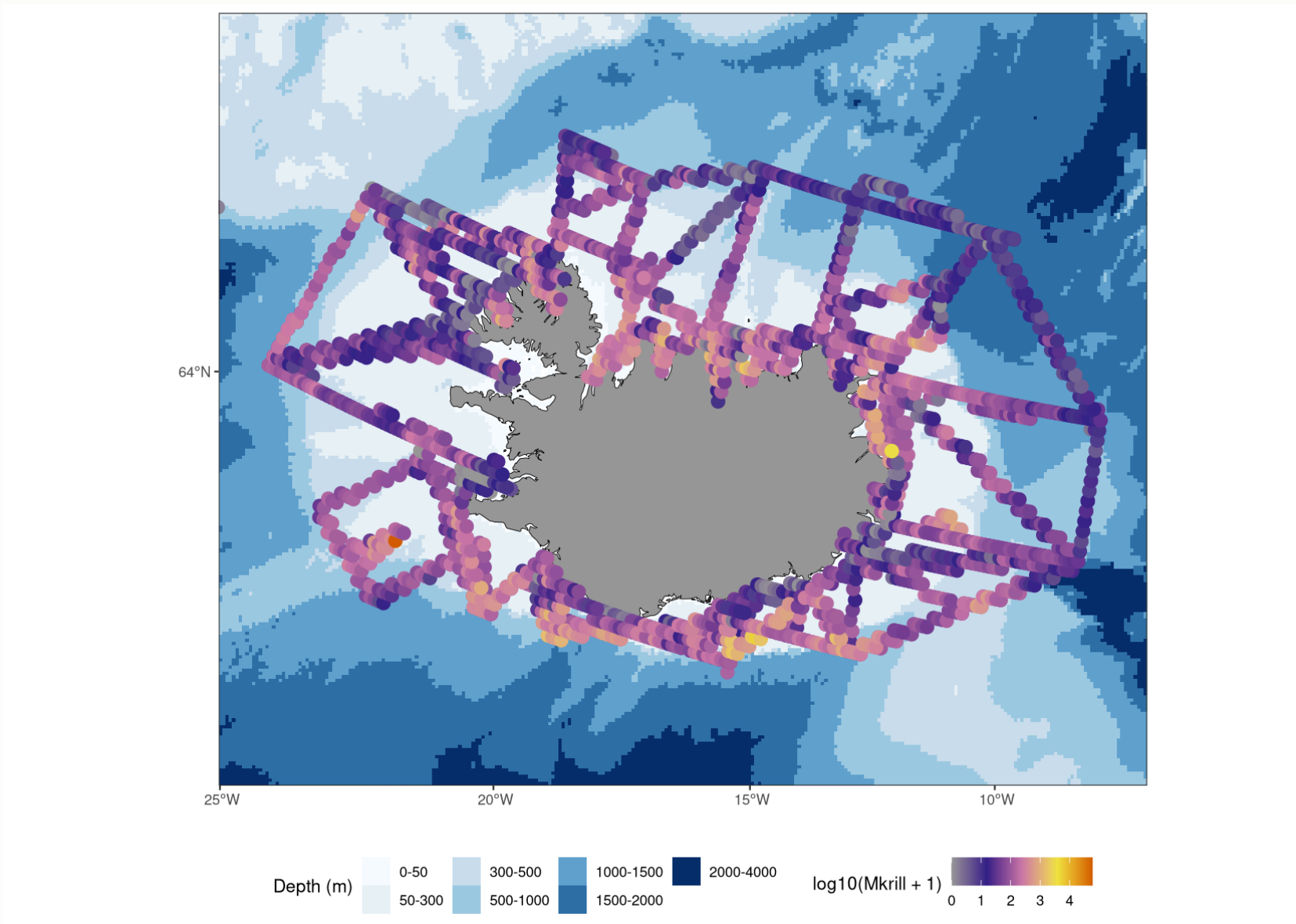
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Korona output in LSSS: backscatter categorised as krill is shown in magenta

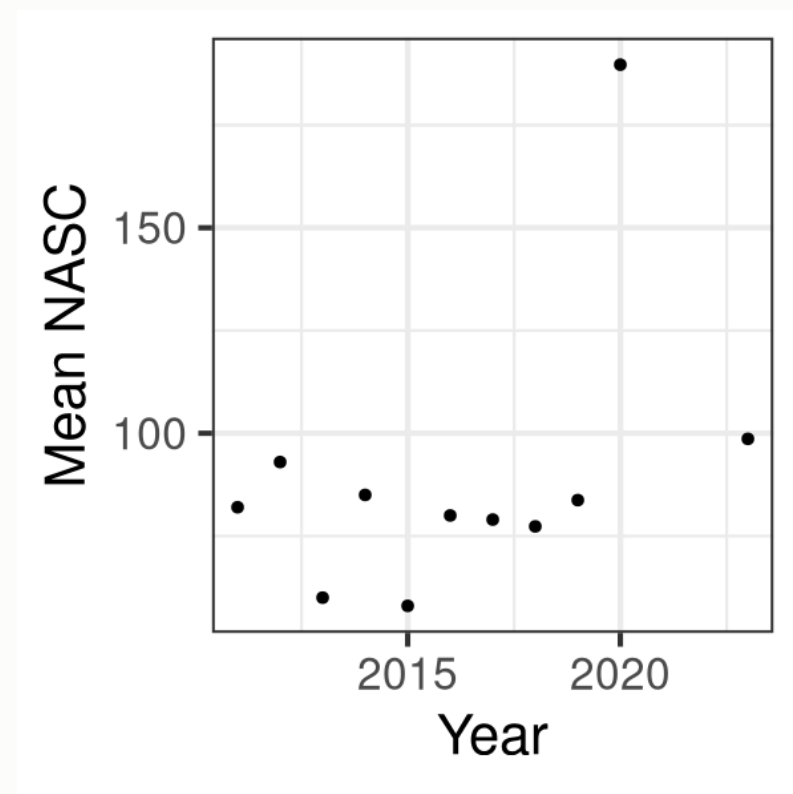
Preliminary findings, 2011–2023

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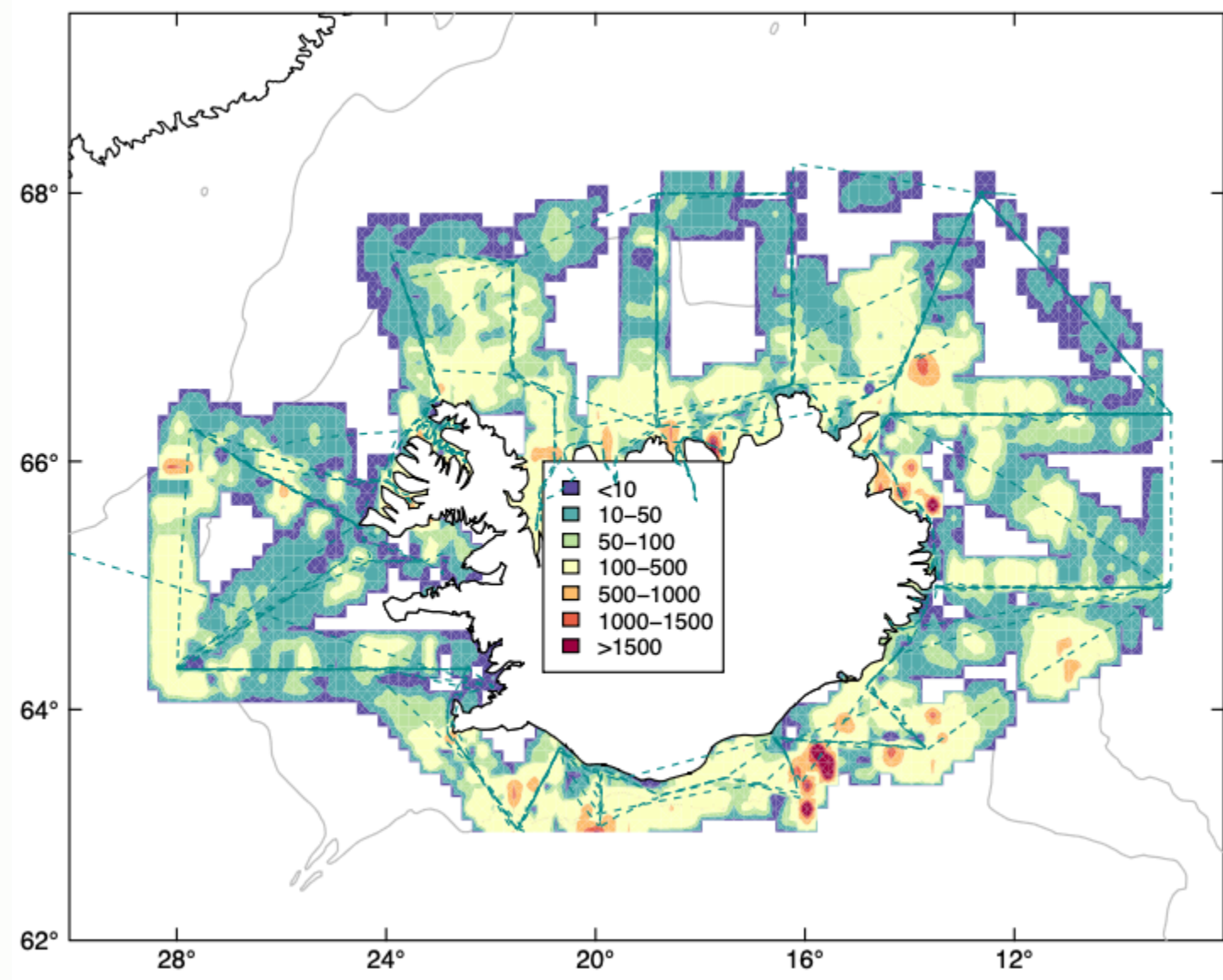
m² nmi⁻²: overall mean krill NASC



Annual mean NASC used as an abundance index. The high value in 2020 reflects extensive coverage in the fjords that year.

Krill distribution

03 · CASE STUDY 1 · KRILL AROUND ICELAND



Average NASC, 2011–2023

Krill backscatter averaged on a grid around Iceland. Colour classes range from below 10 to above 1500 $\text{m}^2 \text{nmi}^{-2}$.

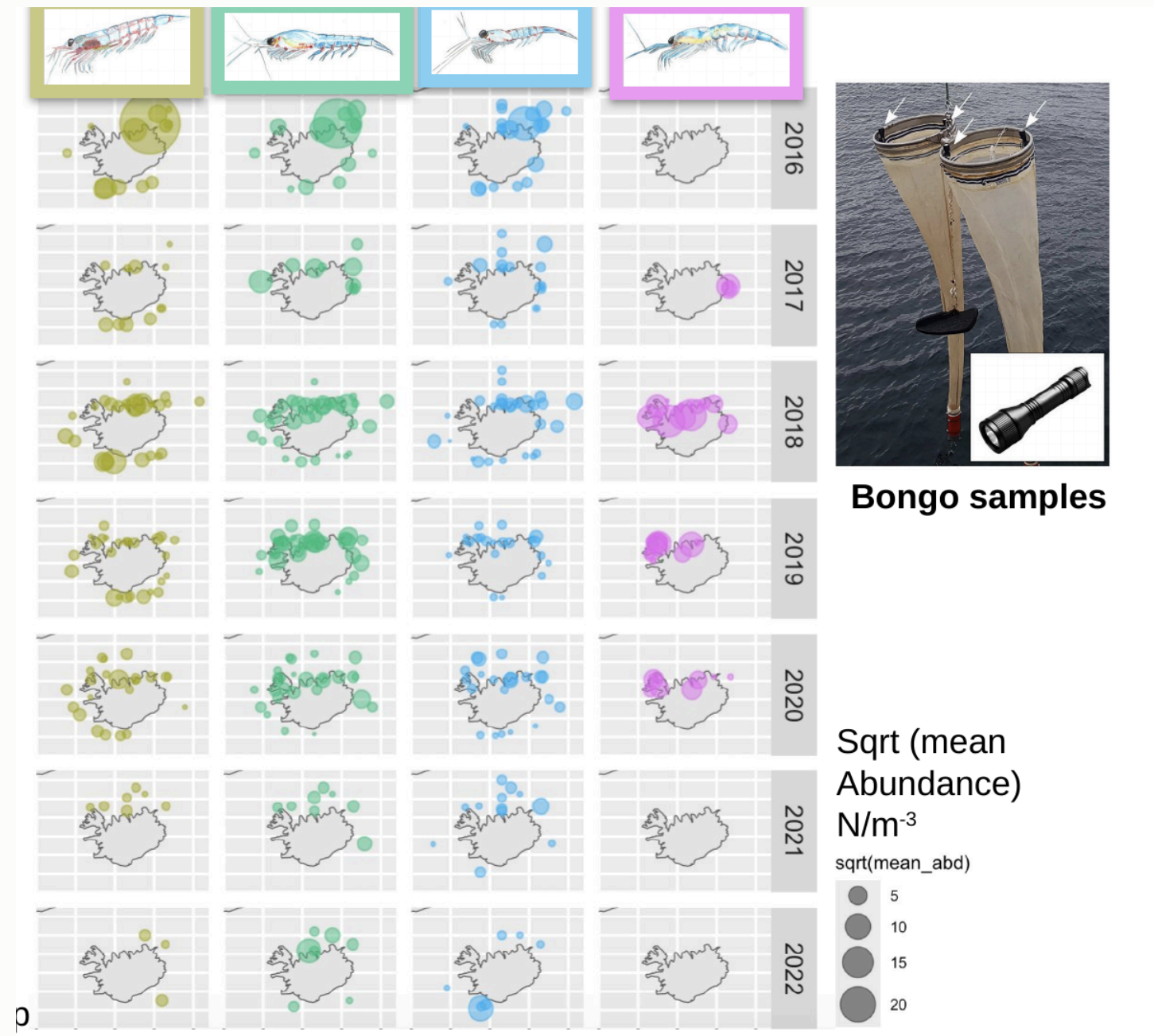
Why it matters

- Shows where krill occur consistently across years
- Basis for comparison with predators such as capelin and baleen whales
- Points to areas where net sampling is most needed

Silva T., Pétursdóttir et al. (in prep.)

Krill species distribution

03 · CASE STUDY 1 · KRILL AROUND ICELAND



Acoustics measure total krill backscatter but cannot tell the species apart. **Bongo net samples** show which euphausiids contribute.

Meganyctiphanes norvegica

Thysanoessa inermis

Thysanoessa longicaudata

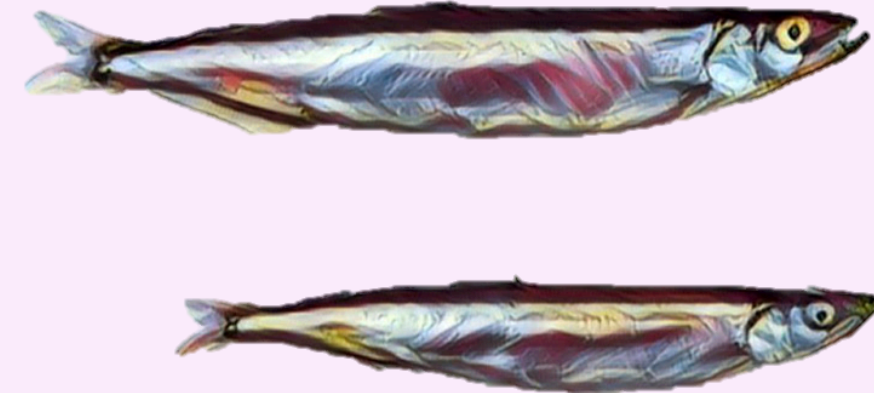
Thysanoessa raschii

Circle size: $\sqrt{(\text{mean abundance})}$, ind. m⁻³. Columns show species, rows show years 2016–2022.

Krill and capelin in East Greenland

03 · CASE STUDY 2

Spatial overlap, water masses and diel vertical migration during the autumn survey



Background and aims

03 · CASE STUDY 2 · EAST GREENLAND

Knowledge gap

Little is known about how krill and capelin interact, or how both are associated with different water masses, in East Greenland during autumn.

Aims

1. Relate the krill–capelin interaction to different water masses
2. Describe the diel vertical migration of both species
3. Measure the spatial overlap of krill and capelin with acoustics

Singh W., Guðnason K., Jónsson S.P., Silva T. (2023)

RELATED PUBLICATION

Rev Fish Biol Fisheries (2026) 36:7
<https://doi.org/10.1007/s11160-025-10019-y>

RESEARCH



Habitat characterization of capelin along East Greenland: insights from hydrographic conditions and feeding distribution patterns

Warsha Singh · Erika Giorgi · Sigurður P. Jónsson · Teresa Silva · Kjetil Våge

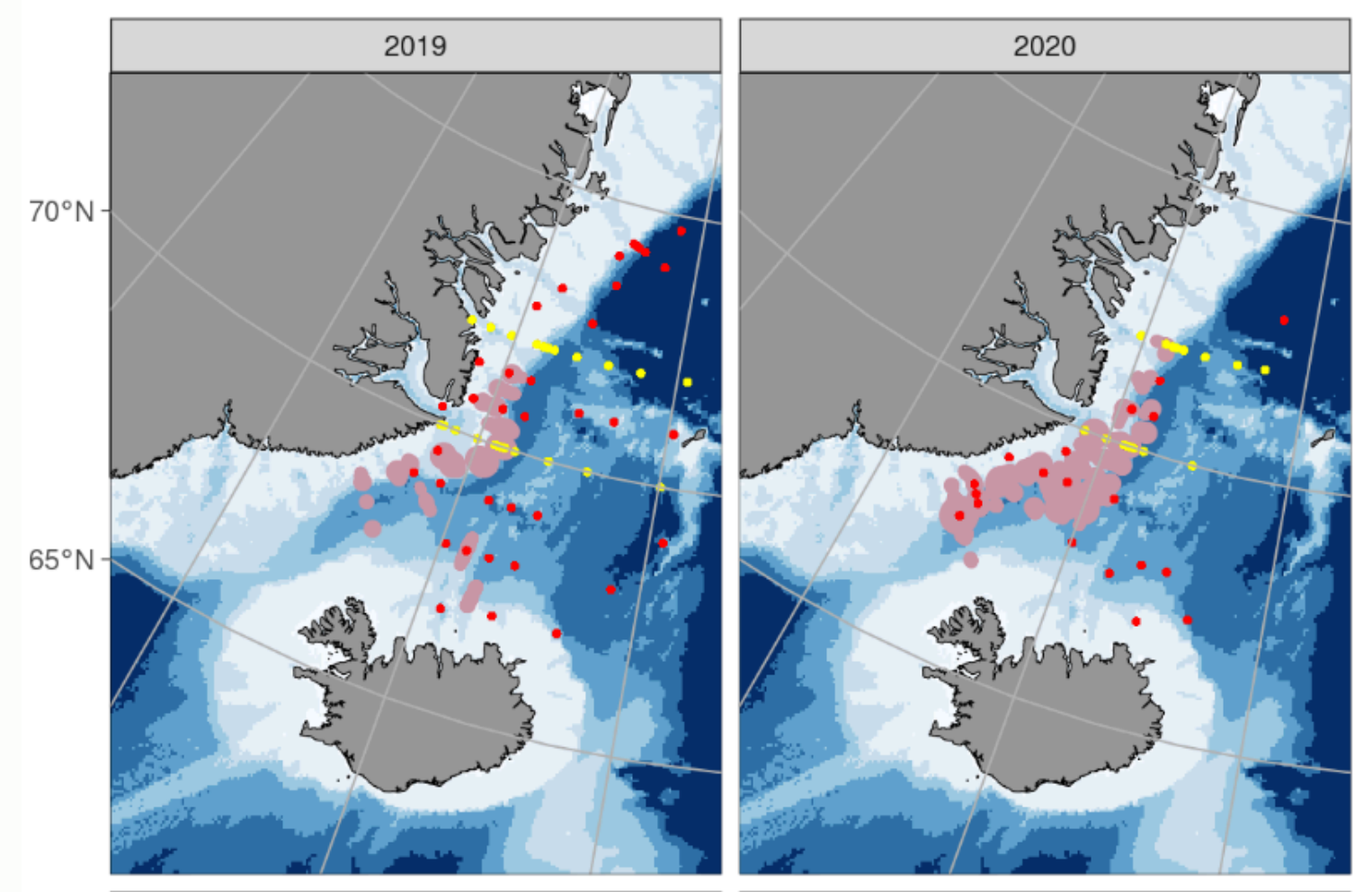
Singh W., Giorgi E., Jónsson S.P., Silva T., Våge K. (2026). *Rev. Fish Biol. Fish.* 36: 7

Data and water masses

03 · CASE STUDY 2 · EAST GREENLAND

Capelin and krill backscatter and CTD profiles were collected on the annual autumn acoustic surveys in East Greenland. Water masses were defined following Håvik et al. (2017) and Rudels et al. (2002).

Code	Water mass	Properties
AW	Atlantic-origin Water	Warmer, higher salinity
PSW	Polar Surface Water	Cold (PSWc) and warm (PSWw) components
PIW	Polar Intermediate Water	Low temperature, relatively high salinity

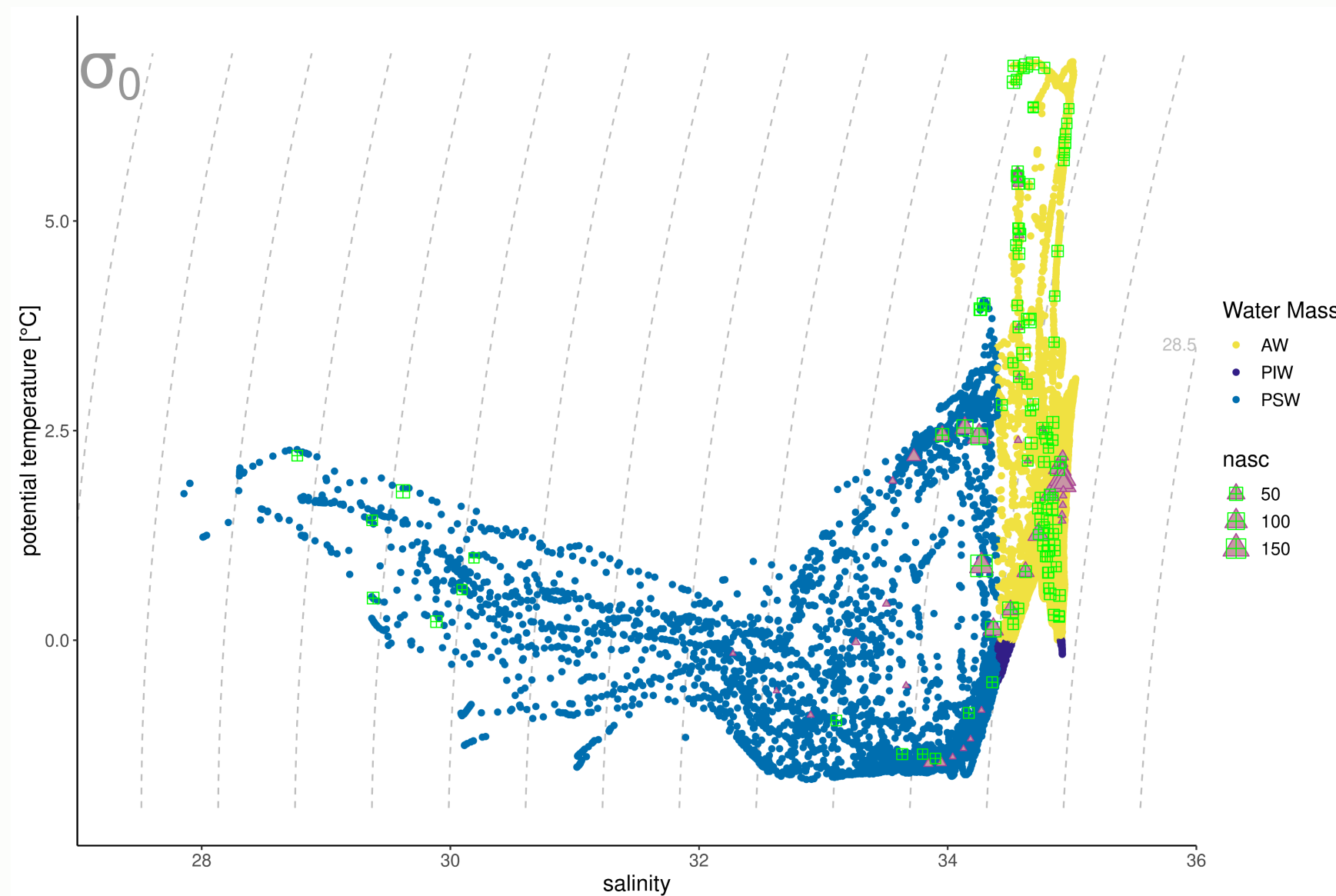


Pink: capelin backscatter · Yellow: CTD stations

Singh W., Guðnason K., Jónsson S.P., Silva T. (2023)

TS diagram, 2019: whole survey area

03 · CASE STUDY 2 · EAST GREENLAND



Potential temperature against salinity for all CTD casts, with acoustic observations of **capelin** (purple) and **krill** (green) placed on the water masses AW, PSW and PIW.

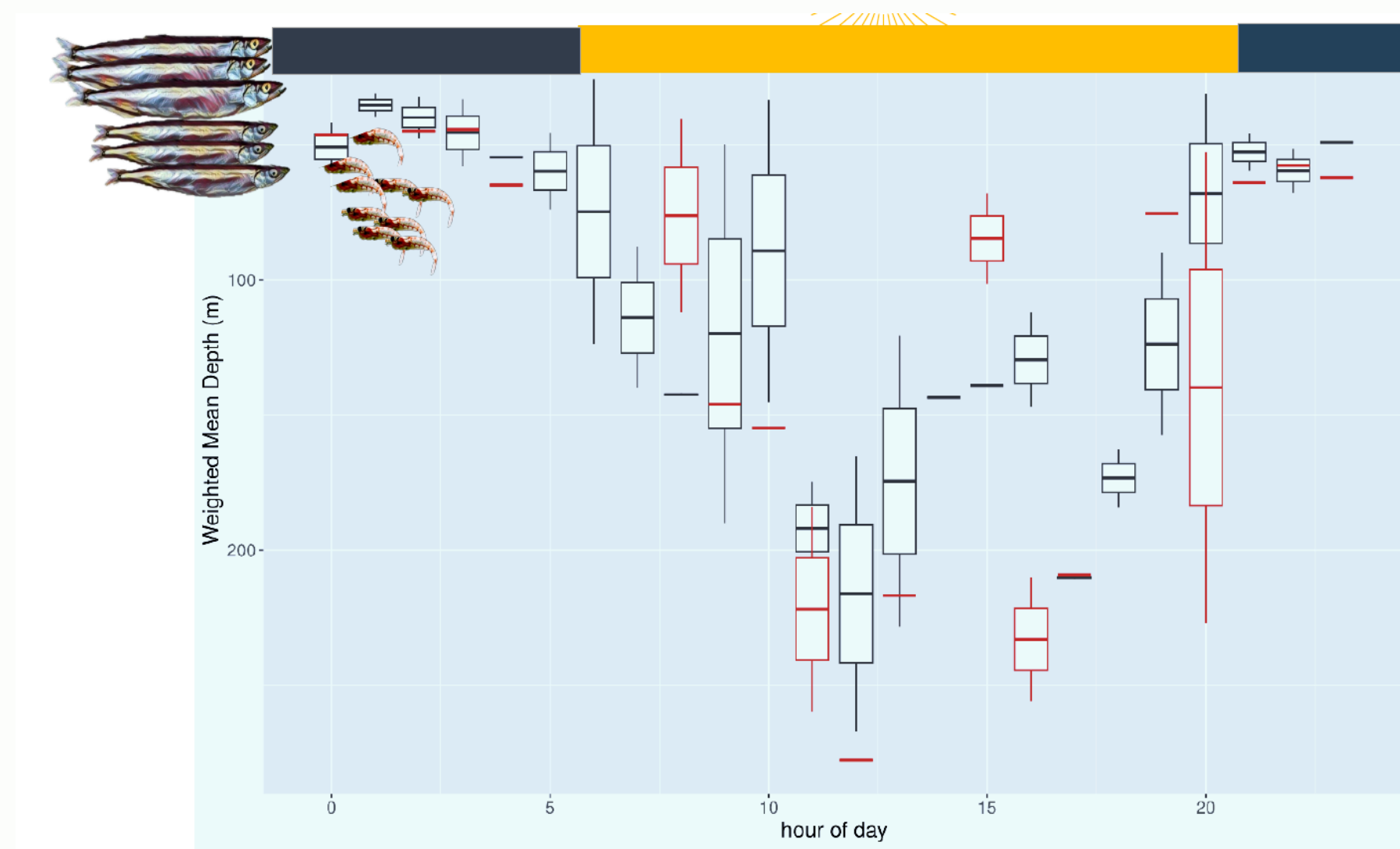
Result

Capelin and krill overlap closely, but krill are more widely distributed. The same pattern was seen in 2020.

Singh W., Guðnason K., Jónsson S.P., Silva T. (2023)

Diel vertical migration

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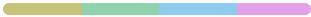
Weighted mean depth by hour of day. Daytime was taken as 08:00–21:00 (late September).

- **Capelin** (black): upper 0–50 m at night, deeper by day
- **Krill** (red): a similar daily pattern

Capelin most likely follow their prey through the water column, crossing temperature and salinity gradients.

Singh W., Guðnason K., Jónsson S.P., Silva T. (2023)

Korona module: hands-on workshop



04 · HANDS-ON

1. **Basic IMR multi-frequency scrutiny**

Start from the standard multi-frequency workflow developed at IMR

2. **Categorisation module: add new categories**

Define a new category and the acoustic features that describe it

3. **Process a new survey**

Run Korona and add NASC values per category automatically

4. **Plankton modules in Korona**

Separate plankton groups using their frequency response

Always double-check the Korona output.

BEFORE YOU START

When in doubt, it is best to exclude the region.

References

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